11 A planet has mass M and radius R.

An object of mass m is moved by means of an external force from the surface of the planet to $3R$ from the centre of the planet.

What is the work done on the object by the external force?

A $-\frac{2GMm}{3R}$ **B** $-\frac{4GMm}{3R}$ **C** $\frac{2GMm}{3R}$ **D** $\frac{4GMm}{3R}$